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Manvendra Rajpoot

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Experience

Software Engineer **CloudTechTiq** Nov 2024 – Present

- Architected a multi-tenant hosting and billing/ERP SaaS on **Django REST Framework** spanning 25+ apps and 230+ ViewSets under a unified `/api/` namespace.
- Designed a dual-gateway payment system (**Razorpay** + **Stripe**) with HMAC-SHA256 webhook idempotency to prevent duplicate charges, coordinating payment, invoicing, provisioning, and double-entry ledger reconciliation, plus Indian GST e-invoicing (EINV-2.1) through the NIC gateway with IRN generation and signed QR codes.
- Extracted billing into a standalone **FastAPI** microservice (billing-ms) as the platform's single billing authority: double-entry ledger, multi-invoice payment allocation and refunds, proforma –> final invoice posting, and multi-entity issuance that resolves legal entity, currency and GST/VAT from the billing address.
- Built an OAuth2/OpenID authorization server and **FastAPI** gateway to **Apache CloudStack** for VMs, volumes, networks and templates, with webhook-first state sync, a polling fallback, and live job progress streamed to clients over **SSE**.
- Built the customer and admin console in **React** 18 + **Redux Toolkit** and Vite across 35+ feature modules (billing, orders, CloudStack, tickets, roles, tax and payment settings), with react-hook-form validation and ApexCharts dashboards.
- Secured the platform with RS512 **JWT** auth (kid-keyed JWKS, refresh-token rotation over HTTPOnly cookies), a 2FA flow over email OTP and RFC-6238 TOTP, and a 43-permission department-scoped RBAC.
- Owned the **Docker Compose** topology for the whole console: 35 services across the Django portal, CloudStack and billing microservices, Celery workers/Beat, PostgreSQL, Redis, **RabbitMQ**, MinIO, NGINX and the observability stack, with a single idempotent `make dev` that rebuilds only on changed build inputs (~4 s warm start).
- Set up end-to-end observability with **OpenTelemetry** tracing and a Prometheus/Grafana/Loki/Tempo stack, so failures across services could be traced to a single request.
- Built typed **TypeScript** API layers with schema-validated contracts (**Zod**/Pydantic), OpenAPI specs and module-boundary checks, keeping frontend and backend in lockstep.
- Built offline-first **React** 19 + TypeScript PWA patterns with TanStack Query and an IndexedDB outbox that replays writes on reconnect, tested with Vitest and **Playwright** end-to-end suites.
- Ran edge and cloud infrastructure: Cloudflare DNS/Pages/Workers, SPF/DKIM/DMARC mail, uptime monitoring, and deployments evolving from a **GCP** VM on single-node **Docker Swarm** with auto-rolling image updates to serverless, weighing cost, limits and operability.
- Planned service migrations and cutovers with written runbooks (a check and a rollback per step), architecture decision records, and a test-parity gate so every existing test is ported or explicitly dropped before release.

Education

Bachelor of Computer Applications (BCA) **Bundelkhand University** Jul 2021 – Jun 2024

- Built an AI-powered news summarizer in Python as a final-year academic project, using the OpenAI API to condense full-length articles into short, readable digests.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, Bash
- **Backend:** Django, Django REST Framework, FastAPI, Hono, Celery, SQLAlchemy, Drizzle, Pydantic, Zod, WebSockets, SSE
- **Frontend:** React, Next.js, Redux Toolkit, TanStack Query, Vite, Astro, Tailwind CSS, Radix/shadcn, PWA/IndexedDB
- **Data:** PostgreSQL, Redis, MySQL, MongoDB, Cloudflare D1/R2, RabbitMQ, Alembic
- **DevOps & Cloud:** Docker, Docker Compose, Docker Swarm, Kubernetes, GitHub Actions, GHCR, NGINX, Gunicorn, Linux/systemd, AWS (EC2, S3, RDS, Lambda), GCP, Azure, Cloudflare (Workers, Pages, DNS), Apache CloudStack
- **Security & Payments:** OAuth2/OpenID, JWT/JWKS, WebAuthn/passkeys, 2FA/TOTP, RBAC, HMAC, Stripe, Razorpay
- **Observability & Testing:** OpenTelemetry, Prometheus, Grafana, Loki, Tempo; pytest, Vitest, Playwright, CI/CD